

An automated approach to multimodal learning analytics of a collaborative making task

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Abstract. With the growth of inexpensive prototyping tools, making based learning has emerged as an important learning paradigm at all levels of education. However, evaluating how conceptual understanding emerges through making-based learning, and when making behaviours are likely to be productive for conceptual understanding remains an analytical challenge. In this work, we contribute a fully automated multimodal learning analytics pipeline that leverages computer vision and machine learning models to extract, from participants audio and video, theoretically grounded behavioural features of making-based conceptual learning as they collaboratively assemble an artefact. By relating these features to students learning gains on a concept test, we can obtain an understanding of behaviours that underlie conceptual understanding. We present one approach based on clustering the behavioural features and relating the behavioural profiles represented by each cluster to students learning gains. This work is one of the first attempts to automate the analysis of and build profiles of productive and unproductive behaviours for collaborative making-based conceptual learning.

Keywords: Making-based learning · collaborative learning · multimodal learning analytics.

1 Introduction

Making, often done in makerspaces, is an interdisciplinary practice fuelled by the rising ubiquitousness of low cost mechanical and electronic tools. It can generate a shift towards the “do-it-yourself” and “innovation” mindsets, and facilitate the learning of concepts and skills [20]. For this reason, making and learning in makerspaces has recently received significant attention within engineering and STEM education, with several institutes incorporating makerspaces and making-based courses in their curricula, where students realize their ideas through prototyping, experimentation, mental and computer simulation, formal and informal knowledge integration, collaboration and reflection [3]. The process of making-based learning, like many other open-ended learning activities, is non-linear and iterative, and so performance alone is an inadequate indicator of students’ learning [11, 5]. Therefore, for pedagogical reasons, it is crucial to evaluate the making-based learning processes of students, rather than their outputs alone.

Making is a complex activity underpinned by recent cognitive science theories [3] which argue that cognition is partially constituted by or dependent on bodily and environmental processes, and the ability to act. Understanding the learning processes of this activity therefore requires a multimodal approach, which considers not merely makers’ actions, but their speech, eye movements, collaboration, prototypes and final products, and their own reflections on the making process. Such an analysis remains a challenge [17] and is typically done using time and effort intensive qualitative methods [23]. Multimodal learning analytics (MMLA) presents an opportunity to make the analysis of learning processes in makerspaces more efficient [17]. However, capturing multiple sources of data reliably using special sensors in a makerspace setting where students are actively working with their hands, collaborating with their peers and potentially moving, is difficult and expensive [17]. We make a case for analysing video and audio recordings of students working in makerspaces, using computer vision and machine learning models to extract the multimodal behaviours mentioned above, and inferring patterns of behaviour, as a promising fully automated approach to multimodal analytics of making-based learning.

Concretely, we investigate this research question: *How can we computationally extract multimodal behavioural features from audio and video recordings, and build multimodal behavioural profiles of conceptual learning in collaborative making-based learning scenarios?* We answer this question by constructing and implementing an MMLA pipeline; we define and extract learning sciences theory inspired multimodal behavioural features of collaborative making-based learning from audio and video recordings of students as they collaboratively work on a making task designed to foster conceptual understanding. We then use these features to build profiles of students who learn and do not learn from the activity. These findings, and further analyses of the multimodal behavioural features, could potentially support the development of AI-powered agents which can serve as making-based learning facilitators by evaluating whether students behaviour is likely to lead to learning, and intervene appropriately if not.

2 Literature Review

Making-based learning is grounded in embodied and collaborative learning theories [3, 21] as students collaborate with peers and educators to make artefacts in a shared space. [3] propose a cognitive model of making-based learning as an iterative process of prototyping, mental simulation, and physical experimentation. Literature also indicates that making-based learning is determined by the nature of the activities, the environment, collaborations and facilitators [10]. Together this suggests that making-based learning happens through the process of trying to reach a solution, and various factors impact this learning process, including students making actions, speech, eye gaze and affect [2]. It is hence critical to evaluate all these factors while assessing students’ learning processes.

Research in making-based learning suggests that a higher proportion of certain groups of “implement” actions (which include undoing parts of a previous

design or putting pieces together) are more productive for learning than other types of actions [25]. Similar research also shows that actions such as meshing, rotating, mounting and making correct connections underlie expertise in making tasks [5]. In that study, participants physical interactions with the objects were used to assess their problem-solving approaches and predict their performance [5]. Finally [15] show that in a making task the quantity and speed of movement of certain hand landmarks differentiates between learners and non-learners. These studies together show how and which physical behaviours can be used to understand students' learning during hands-on work, and based on this, we conjecture that different *types* of actions underpin different learning gains from making activities.

When students learn collaboratively, research suggests that a common ground and shared understanding is necessary for collaboration to be effective for learning [6]. Speech is one of the main modes through which students build this shared understanding. Research has shown that the nature of dialogue such as taking turns and building on each other's reasoning [13], as well as features of speech such as pitch and loudness, can both be indicative of good collaboration [24]. Students are not, however, wholly dependent on language to build and maintain shared understanding. In collaborative settings, shared understanding can be established using eye gaze as well [6]. For instance, features of joint visual attention such as cross-recurrence and gaze similarity have been shown to be associated with better collaboration quality and learning gains [18, 16]. Body postures and/or gestures are another means of building shared understanding in collaborative settings. MMLA has been used to identify markers of good collaboration such as synchronicity in problem-solving tasks [12, 22]. Finally, affect has a crucial role in learning, and researchers have used arousal and valence, which indicate affect, to study the nature of collaboration. In [8], the author found that the process of building an understanding was associated with negative emotions. Similarly, [11] found that learners in a collaborative open-ended task are associated with both positive and negative emotions.

Together the above literature suggests that analysing the learning processes of collaborative making-based learning requires considering individual and collaborative actions or movement, speech, eye gaze and affect together to obtain a complete understanding. While actions and movements of individual making-based conceptual learning have been studied [15, 5] and MMLA has been used to evaluate collaboration in collaborative hands-on learning situations (for instance, [22, 4]), MMLA of conceptual learning in collaborative making scenarios are limited. [25] performed a multimodal analysis of a making activity using audio, movement and electro-dermal activation measured using sensors, along with human-annotated data. However, such sensors can be both intrusive and expensive in a classroom making set-up. In this work, we bridge this gap and contribute an MMLA pipeline to assess collaborative making-based conceptual learning. Specifically, we present a computational approach to build profiles of learners and non-learners using computer vision and machine learning extracted, theory-inspired multimodal behavioural features from video and audio record-

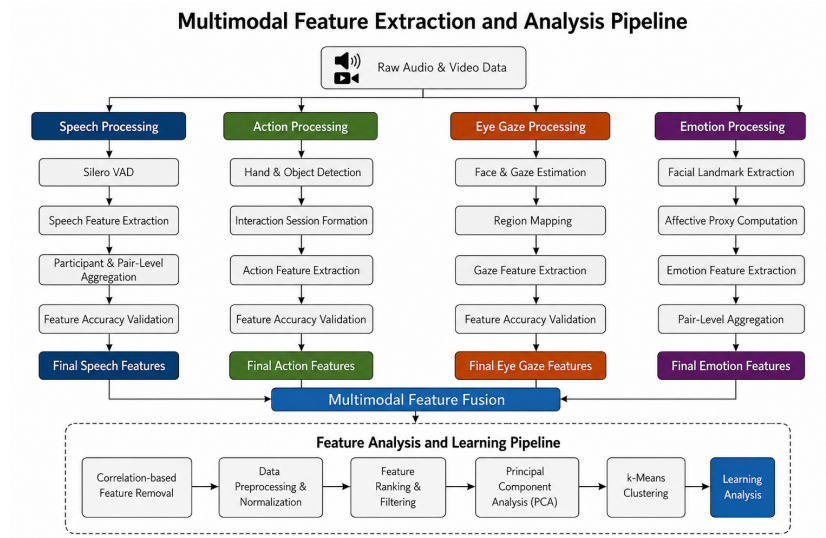


Fig. 1. MMLA pipeline using behaviours

ings. We test this pipeline in a pilot study with 17 pairs of learners and present preliminary, indicative findings of the types of behavioural profiles of learning in a specific making task, enabled by our pipeline. Next we describe our pipeline.

3 MMLA Pipeline for assessing collaborative making-based learning

To capture the nature of the collaborative making-based learning process we propose an MMLA pipeline for relating multimodal behaviours to learning outcomes, shown in Figure 1, and elaborated next.

3.1 Feature extraction and selection

We extract four sets of literature-inspired features to understanding collaborative making-based learning- actions, speech, eye gaze and affect, as explained below.

Actions: We analyze top-view video recordings of participant pairs’ hands in collaborative making tasks to detect and quantify participant interactions with objects, while controlling for issues such as noise, occlusion, temporal fragmentation, hands leaving the frame, and hovering without interaction. We employ a multi-stage, rule-based pipeline integrating hand tracking, object detection, temporal smoothing, and interaction confirmation for stable, meaningful results.

Standard YOLO object detection failed due to task-specific objects; training or fine-tuning a YOLO model to robustly detect all task objects was beyond the

scope of this work. Instead, template-based detection using color and size constraints ensured consistent, accurate, interpretable identification. Hands were tracked with MediaPipe Hands [9] using bounding boxes and centroid-based tracking for identity consistency. Palm positions were temporal smoothed using a sliding-window to reduce noise. In the shared workspace, hands were initially assigned to participants via a fixed diagonal image partition, and stabilized by majority voting over short windows to counter crossings or noise. Per participant, hands received fixed labels (e.g., Hand 1, Hand 2) by horizontal position. Interactions required confirmed hand-object contact with temporal persistence rules, merging touches into sessions with start/end times. From stabilized trajectories and sessions, we derived participant- and pair-level features capturing interaction frequency, spatial coverage, object handling, and collaborative dynamics, focusing on behavioral patterns, not task semantics.

Speech: We began with individual voice activity detection, and speech and silence segments were identified using a neural voice activity detection framework called Silero Voice Activity Detection (Silero VAD [19]) selected for its robustness to background noise, low computational overhead, and ability to detect short speech segments such as brief acknowledgements and interruptions. Further, previous evaluations have shown that Silero VAD outperforms classical energy-based and WebRTC-based approaches across diverse acoustic conditions [19]. For each audio recording, Silero VAD was used to generate a sequence of continuous speech segments represented as start–end time intervals. These timestamps formed the basis for all subsequent speech feature extraction. The analysis focuses on behavioural interaction characteristics rather than the semantic content of dialogue, consistent with the overall philosophy of our multimodal behavioural pipeline. Accordingly, the speech modality characterizes communication through features such as turn-taking, pauses, overlap, pitch, and vocal energy, which have been widely associated with collaboration quality while remaining language-independent and computationally efficient [11, 24]. Furthermore, semantic analysis depends on accurate speech transcription, which remains challenging for spontaneous, overlapping, and code-mixed Indian-English conversations. By relying on behavioural speech features, the proposed pipeline remains scalable and broadly applicable to multilingual educational settings.

Eye Gaze: We use GazeLLE [14], a deep learning-based model designed to predict visual attention in natural images. Accurate eye gaze estimation however requires reliable localisation of the participant’s face in each video frame, for which we use OpenFace [1]. GazeLLE takes a full video frame together with a normalised face bounding box and produces a spatial representation of gaze in the form of a probability heatmap over the image. Unlike traditional approaches that output a single gaze vector or angle, this heatmap representation captures uncertainty and spatial spread in visual attention. Frames with low in/out confidence were excluded to avoid unreliable estimates. For valid frames, a single gaze point is extracted by selecting the peak location of the heatmap. To reduce frame-level noise and short-lived fluctuations, gaze trajectories are temporally smoothed using a moving average filter prior to further analysis.

Each participant is recorded using an individual camera, with the shared workspace directly in front of them, while their collaborator was positioned laterally (to the left or right, depending on the participant, see figure 2b). This fixed spatial arrangement allows gaze direction to be interpreted relative to predefined regions of interest, such as the workspace, the partner, and other areas of the scene. To relate gaze direction to these regions of interest, gaze points are manually mapped to these regions (workspace, partner, and elsewhere) based on the physical task layout and fixed front-facing camera placement. From gaze points, we compute measures of individual and joint visual attention such as number and durations of fixations, gaze shifts, gaze focus spread, shared attention and time spent in predefined regions of interest; individual features are averaged for each pair while joint attention features are retained as is.

Affect: To capture participants’ affect, we extract frame-level facial features and derive continuous affective proxies from them. Facial landmarks are extracted using MediaPipe FaceMesh [9], and rather than storing the complete landmark representation, we compute a compact set of interpretable geometric features derived from selected landmark pairs. These features such as eye openness, mouth opening, smile intensity, eyebrow movement, and lip compression, were chosen based on affective-computing literature associating smile/lip-corner movement with positive valence, brow lowering with negative valence, and eye opening with arousal [26]. Landmark coordinates are normalised per frame and scaled using inter-ocular distance to reduce the effects of head movement and scale variation. To reduce frame-level noise, exponential moving average smoothing was applied. All values were calculated framewise, and averaged for each pair. Valence and arousal were computed as weighted linear combinations of these geometric features, following the continuous valence-arousal representation commonly used in affective computing. The specific weighting coefficients are heuristic: they were fixed a priori, before any downstream analysis, and were not learned, tuned, or optimised on our dataset. Accordingly, these measures should not be interpreted as clinically validated or empirically calibrated affect scores, but as simple, interpretable behavioural proxies intended to support relative comparison and clustering across participants.

3.2 Building profiles of Making-Based Learning

The extracted features, which can be aggregated across the entire task or computed for smaller time segments, can now be used to develop an understanding of the collaborative making-based learning processes using appropriate machine learning algorithms. Here we propose an analysis based on [11] where the aggregated multimodal features are clustered, and the resulting clusters’ learning gains and feature values are statistically compared to draw inferences about behaviours associated with learning and not learning. Before clustering, we perform a correlation analysis to identify redundancy among the features. Highly correlated features (Pearson’s correlation > 0.7) are removed to minimise multicollinearity. Next we perform feature selection using appropriate methods, for

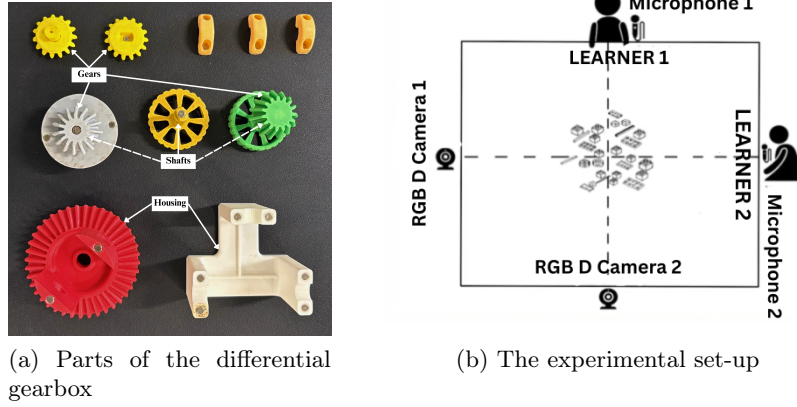


Fig. 2. The making task and experimental set-up

instance, using an “information gain” metric to evaluate the gain in information about learning vs not learning obtained when using a feature, and choosing the top 5 features. Finally, we perform K-means clustering using this subset of features, choosing an appropriate K according to the silhouette or elbow method.

4 Methods

To evaluate our MMLA pipeline, we designed a pilot study where pairs of students worked on a making task as detailed below.

4.1 Making Task, Instrument and Participants

The task involved assembling a differential gearbox from its parts (Figure 2a). Participants were given ten magnet-embedded pieces without any instructions, diagrams, or hints about the final structure. The magnets made the pieces connect together easily, resulting in many incorrect partial assemblies, but only one specific arrangement yielding a fully functional differential gearbox. The task was taken from [5], where it was shown to distinguish between experts and novices, and was selected because it provides an authentic, hands-on making activity directly related to the concepts of gear mechanisms, emphasizing key principles such as gear meshing, torque transfer, and proper alignment, while also ensuring consistency through standardized materials and a controlled workspace.

To assess participants’ conceptual understanding before and after the making activity using 10-item multiple-choice pre- and post-tests. These tests had textbook style questions such as *When one shaft transfers motion to another shaft placed at an angle, what is the key factor that determines how the motion is transferred?*, designed to capture students understanding of all differential gear

mechanisms concepts mentioned above. Two matched versions, Test A and Test B, were created by an experienced Mechanical Engineering instructor, tested with a small set of participants and revised based on the findings. Within each dyadic session, one participant received Test A as the pretest and the other received Test B, with the tests swapped at posttest, ensuring counterbalancing and minimizing potential test-related bias.

Thirty-four participants in dyads (17 pairs) participated in the study. The participants were recruited from within the student community of our institute, following the guidelines of the Institutional Ethics Committee. Participants were second or third year Engineering undergraduates. Written informed consent was obtained from all participants prior to the experiment. The participants knew one another prior to the study, as they were instructed to select their own partners, thereby minimizing time spent on familiarization and ensuring they were comfortable during the collaborative task.

4.2 Experimental setup and protocol

The task was recorded using a high-quality camera and speech acquisition system, as follows. One Brio webcam, placed at the top, captured the workspace and participants' actions, including the hand gestures used to perform the task. Sensitive Lavalier microphones attached to each participant were used to capture speech data. Two Intel RealSense cameras were placed one in front of each participant to capture their faces and part of the workspace. The participants were seated at a 90-degree angle to each other as shown in Figure 2b, to maximize visibility of gaze patterns, facial expressions, and hand gestures. The front cameras were equidistant from each other and the task pieces were placed at the center of the table between the participants.

At the start of each session, we ensured that each participant had a clear view of the workspace. Next each participant completed a pretest individually within 10 minutes, during which they were explicitly instructed not to speak or collaborate. Following this, the participants (in dyads) were given 10 minutes to collaboratively assemble the pieces on the workspace. No other instructions were provided except encouraging participants to communicate freely with each other. The entire session was recorded using the audiovisual setup described earlier, following which each participant given 10 minutes to complete the post-test individually.

4.3 Data Analysis

We implemented the pipeline described in Section 3, first extracting features of participants speech, eye gaze, emotions and actions from video and audio recordings. Where appropriate (detailed in Section 5.1) we evaluated the accuracy of the features and retained only those features which showed accuracy $>70\%$. We then performed correlations and feature selection to retain meaningful features, and finally clustered the reduced subset of features to develop learner and non-learner profiles.

Learning gains (LG) were computed as (Posttest score) – (Pretest score), while normalized learning gain (NLG) [7], signifying the gain relative to the maximum possible gain, was computed as $\frac{\text{Posttest score} - \text{Pretest score}}{10 - \text{Pretest score}}$. The individual learning gains were averaged to obtain the learning gains of each dyad. Pairs with learning gain > 0 were labelled *learners* and those with learning gains ≤ 0 , *non-learners*.

5 Findings

5.1 Feature Accuracy

Our feature extraction resulted in 29 features including 13 speech features, 8 gaze features, 2 affect features and 6 action features.

Actions: The accuracy of action features was evaluated through manual validation on randomly sampled 30-second video segments (5% of the total dataset). These segments, drawn from multiple participant pairs, were manually annotated for interaction events and object switches and compared against the automatically extracted action features for the same intervals. While average interaction duration showed comparatively low accuracy ($< 70\%$) in manual validation, interaction counts (71 %) and object switch counts (76 %) showed good accuracy and were retained for further analysis.

Speech: The reliability of speech activity detection was assessed using a consensus-based evaluation, in which Silero VAD outputs were compared against multiple reference configurations, including WebRTC VAD (aggressiveness modes 2 and 3) and frame-level energy-based detection. Silero VAD achieved a mean precision of 0.938, a recall of 0.743, and an F1-score of 0.809. The high precision indicates a low false positive rate, ensuring that detected speech segments are highly reliable for downstream feature computation. The comparatively lower recall reflects a conservative detection strategy that excludes marginal or low-energy vocalisations. Silero VAD has been previously validated across diverse acoustic conditions [19], and its consistent performance in our evaluation suggests that we obtained reliable speech features in this study.

Eye Gaze: As the regions corresponding to “workspace”, “partner” and “elsewhere” were manually defined, we performed human verification of only the features related to these regions to establish their accuracy. These labels were visually inspected by reviewing slowed-down recordings (for 5% of the data) and comparing them with the automatically assigned gaze contexts. Across the dataset, the average accuracies observed were 76.52% for workspace gaze, 59.35% for elsewhere gaze, and 3.06% for partner-directed gaze. The lower accuracies of the latter two features are primarily associated with the final region-assignment step, where continuous gaze estimates are mapped to manually defined regions of interest to obtain contextual gaze labels. Consequently, small variations in the estimated gaze location may result in different contextual labels, particularly near region boundaries, without necessarily indicating inaccuracies in the underlying gaze estimation process. These region-based features represent a contextual interpretation of the estimated gaze locations and hence the two features

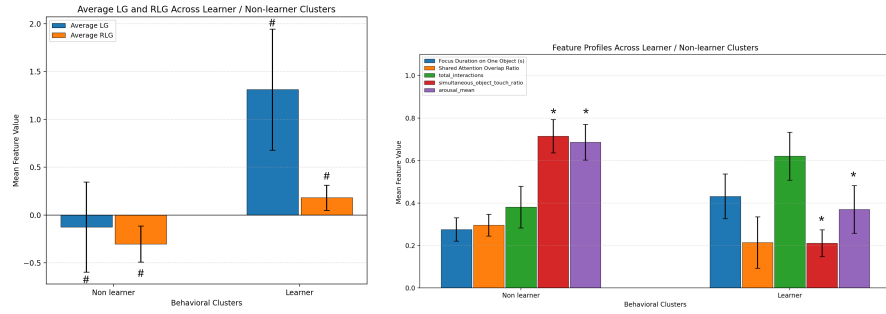
that did not satisfy the 70% accuracy criterion were excluded from further analyses. The remaining eye-gaze features were computed directly from the temporally smoothed continuous gaze trajectories and probabilistic gaze heatmaps produced by the model using deterministic feature computation, and as such their accuracy relies on the accuracy of the GazeLLE model itself.

Affect: MediaPipe FaceMesh has been shown to provide accurate and robust facial landmark estimation across varying lighting conditions, head poses, and facial expressions, making it suitable for behavioural analysis in unconstrained settings [9]. Since the study relies on geometric features rather than categorical emotion labels, no additional accuracy evaluation was performed.

5.2 Feature Selection and Clustering

After accuracy evaluation, our feature set came down to 26. After correlational analysis we obtained 14 uncorrelated multimodal behavioural features, listed below, that can be used to understand making-based learning, and encompass both individual and collaborative behaviour. We further performed feature selection using information gain, retaining the top 5 features for clustering namely, *Focus Duration on One Object (s)*, *Shared Attention Overlap Ratio*, *total_interactions*, *simultaneous_object_touch_ratio* and *arousal_mean*. Finally we performed K-means clustering based on these 5 features and selected 3 clusters as this gave the maximum silhouette score. One cluster had only one element and while interesting, this was ignored for further statistical analyses.

1. Physical Exploration
 - (a) *total_interactions*: Number of confirmed hand-object interactions during the task.
 - (b) *object_switch_rate_per_min*: Rate of transitions between different objects across consecutive interactions, normalised per minute.
 - (c) *workspace_coverage_ratio*: Proportion of the workspace explored, computed from the convex hull of temporally smoothed palm positions.
 - (d) *simult_object_touch_ratio*: Proportion of touch events in which both participants simultaneously touch the same object.
2. Affective State
 - (a) *arousal_mean*: Continuous proxy for emotional activation ($0.55 \times \text{eye_open} + 0.45 \times \text{mouth_open}$).
 - (b) *valence_mean*: Continuous proxy for emotional positivity ($0.65 \times \text{smile_intensity} - 0.25 \times (1 - \text{brow_frown}) - 0.10 \times \text{lip_press}$).
3. Verbal Communication
 - (a) *overlap_duration_s*: Sum of time intervals where both participants speak simultaneously.
 - (b) Total turns: Count of non-overlapping speech transitions from one participant to another
 - (c) *avg_pitch_Hz*: Average pitch across all speech frames across both participants.



(a) Learning gains of learner and non-learner groups * indicates $p < 0.05$, # indicates $0.05 < p < 0.1$ (b) Learner and non-learner collaborative making-based learning profiles. * indicates $p < 0.05$, # indicates $0.05 < p < 0.1$

Fig. 3. Learning and behavioural profiles

(d) avg_energy: Mean-squared signal energy across speech segments across both participants.

4. Visual Attention

- (a) Focus Duration on One Object (s): Average duration for which a participant continuously maintains gaze on the same spatial location
- (b) %Time Looking at Workspace: Percentage of time the participant's gaze is directed toward the shared workspace
- (c) Average Focus Area Spread: Average spatial dispersion of gaze points during fixations
- (d) Shared Attention Overlap Ratio: Degree of spatial overlap between participants' gaze heatmaps over time

The mean values of the learning gains and multimodal features of each cluster are shown in Figure 3. We find that one cluster has an average LG and NLG greater than zero, and is hence considered a learning cluster, while the other cluster with average LG and NLG less than zero is considered a non-learning cluster. We observe that learners have significantly lower *simultaneous_object_touch_ratio* and *arousal_mean*. This suggests that pairs who learned had fewer instances of simultaneously handling the same object and they were calmer throughout the activity. Learners also interacted more with the objects, spent a longer time focused on one object and lower time periods of looking at the location, though these differences were not statistically significant ($p > 0.05$, with medium-high effect sizes) possibly due to the small sample size. Together this indicates a collaborative learning profile marked by increased interactions, a longer individual gaze focus, looking at different parts of the workspace and handling different objects, and a calmer emotional state.

6 Discussion and Conclusions

In this paper, we presented a video and audio-based fully automated MMLA pipeline to characterize behavioural patterns of learning processes during a collaborative making-based task. Grounded in embodied and collaborative learning theories, the study identified reliable multimodal features of learners’ actions, speech, eye gaze, and affect that can act as indicators of learning during collaborative making. The main contribution is an MMLA pipeline that can generate a set of accurate features derived from multiple modalities, and that represent both individual and collaborative aspects of making-based learning (see Section 5.2 for the list). For making actions, our final feature set capturing both participant- and pair-level features, includes interaction counts, workspace coverage, object handling frequency, and simultaneous object handling. These features align with prior work suggesting that specific patterns of physical action, quantity and speed of movement are central to productive engagement and expertise in making tasks [25, 5, 15]. Our results suggest that such action features can be reliably captured using vision-based approaches in a collaborative setting, without the need for wearable sensors.

Speech features achieved high precision, and our final set included features of individual and collaborative speech such as speech overlap, turn taking, energy and pitch shown in prior research to be indicative of the quality of collaboration and learning [24, 13, 11]. Our findings are consistent with this literature in indicating that such speech features can be reliably extracted from audio data. The final set of eye gaze features include measures of individual (focus duration, time looking at workspace, focus area spread) and joint attention (shared attention overlap) aligning with previous research which suggests that both sets of measures are related with collaboration quality and learning [18, 16] and we find that such measures can be reliably captured using a deep learning model on video recordings. For affect we relied on geometric facial features, and MediaPipe FaceMesh has been shown to provide accurate results in multiple contexts [9]. This approach aligns with prior research such as [11, 8] that has integrated affective dimensions, such as valence and arousal, to assess collaborative learning. Together with action, speech, and gaze features, these affect-related measures contribute to a more complete understanding of collaborative making-based learning.

The secondary contribution of this work is to build one possible multimodal behavioural profile of collaborative making-based learning. We cluster a subset of 5 multimodal features, and obtain two clusters which show differences in their learning and behaviours. Concretely, learning teams show more interactions and more visual focus on objects than non-learning teams. Importantly, learning teams visually and physically work in separate parts of the workspace. While this finding contradicts previous research on the importance of joint visual attention [18, 16] in learning, this behaviour may be especially important in making to allow each student to interact with and observe the objects separately, and merits further investigation. Finally, learners have a calmer affective state, agreeing with literature from MMLA of open-ended tasks and making [11, 25]. One interesting

finding was that our participants spoke very less (an average of 1.5 minutes in a 10 minute task). This could be because we deliberately did not script the collaboration to remain authentic to making tasks, and suggests that students may have communicated through their actions on the object rather than speech, a finding that is worth further exploration.

This study has certain limitations. First, the small sample size limits the stability of the clustering and generalization of the preliminary behavioural profiles. While we did obtain significant differences on certain features, the lack of statistical differences in other features suggests that our sample size could be insufficient to detect the effects. Second, our findings apply to a specific type of task, namely the differential gearbox assembly which requires specific kinds of interactions that may not extend to other tasks. Nonetheless, the primary goal of this work was not to build the behavioural profiles itself, but to present a reproducible and fully automated MMLA pipeline for collaborative making-based learning that can allow building behavioural profiles. It lays the foundation for future research in more ecologically valid, collaborative making-based learning environments. Semantic analysis of speech and actions represents an important direction for future work.

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